

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Second Semester of M. Tech (IT) Examination****May 2018****IT765 Internet of Things****Date: 08.05.2018, Tuesday****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the following questions. [07]**

- a. What is RFID?
- b. What is wireless sensor node?
- c. What is ubiquitous networking?
- d. What are the internal components of IoT device?
- e. What is streaming analytics?
- f. What do you mean by context aware computing?
- g. What is Near Field Communication?

Q – 2.a What is smart city? What are the positive implications of building smart city? [05]**Q – 2.b Answer any two questions. [08]**

- (i) Explain the working of smart hospital.
- (ii) Explain 6LowPAN protocol stack.
- (iii) Explain complex event processing with block diagram.

Q – 2.c List and explain any three sensors. [03]**Q - 3 Answer any two questions. [12]**

- a. Explain IoT-A Reference Model.
- b. Draw and explain the block diagram of ETSI M2M architecture.
- c. What is MQTT? How MQTT helps in IoT?

SECTION – II**Q - 4 Answer the following questions.**

- a. Why privacy and security is a major concern for IoT? [01]
- b. With IPv6, there are approximately _____ unique addresses can be formed. [01]

1. 3.4×10^{38}	2. 3.4×10^{37}
3. 3.6×10^{38}	4. 3.5×10^{37}

- c. What is wafer chip manufacturing technology? [02]
- d. What is the impact of Ransomware on IoT devices? [02]
- e. What is software defined radio? [01]
- Q – 5.a** What is real time analytics? Explain real time analytics with its applications. [05]

OR

- Q – 5.a** Explain various methods of predictive analytics approaches. [05]
- Q – 5.b** What are the research issues in IOT? Explain any one. [03]
- Q – 5.c** List and explain IoT wearable devices. [05]
- Q – 6.a** What is cloud computing? Explain how cloud computing helps in IoT? [07]

OR

- Q – 6.a** Draw and explain the block diagram of I Core Architecture. [07]
- Q – 6.b** **Answer any two questions.** [08]
- (i) Explain cross domain IoT applications.
- (ii) Why interoperability is a measure challenge for IoT?
- (iii) Compare gateway and network layer technologies.
